Mohit Dubey

AI Full Stack Engineer

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SUMMARY

Senior Full Stack / AI Engineer with 5+ years building production **Python** backend services and **Generative AI / Agentic AI** systems end to end: multi-agent orchestration, planning, tool-calling, and memory/state management (**LangChain**, **LangGraph**, **DeepAgents**). Hands-on with **RAG** architectures (embeddings, vector databases, chunking, retrieval optimization), **MCP** tool-integration standards, low-code/configuration-driven agent platforms for business users, and **FastAPI** and **asyncio** services with SSE/WebSocket streaming, resilience patterns, and structured observability. Experienced with **Kubernetes** and **OpenShift**, **MongoDB** for state/checkpointing, **CI/CD** (**GitHub Actions**, **Jenkins**, SonarQube), and enterprise auth (AD/SSO, RBAC, OAuth2).

WORK EXPERIENCE

Software Engineer —— Publicis Sapient June 2025 - Present

- Led architecture of an enterprise-grade **GraphRAG**-based private coding assistant using **LangGraph**, **DeepAgents**, and agentic AI, improving retrieval accuracy by 38% and cutting developer resolution time by 60%.
- Designed a **low-code, configuration-driven agent platform** letting non-technical business users configure, deploy, and manage AI agents from reusable templates.
- Built an **MCP server** with dynamic tool discovery, semantic search, and context orchestration to connect agents with enterprise systems/APIs, cutting LLM context overhead by 70%.
- Integrated **LiteLLM** as a unified LLM gateway with cost-based fallback, rate limiting, and centralized usage tracking across agent workloads.
- Orchestrated **multi-agent** planning, tool-calling, and memory with **MongoDB**-backed state and checkpointing, **SQL Server** and **Oracle** integrations, guardrails, and policy enforcement for secure multi-tenant deployment.
- Architected **RAG** pipelines on **LangGraph** with embeddings, vector databases, chunking, and hybrid retrieval/reranking, increasing answer relevance by 42% for thousands of concurrent enterprise requests.
- Authored **PL/SQL** stored procedures, packages, and functions on **Oracle** for enterprise data access used by agent tools, with query tuning and indexing for fast responses under load.
- Exposed agent capabilities via **FastAPI**, **Flask**, and **asyncio** with SSE/WebSocket streaming, circuit-breaker/retry resilience patterns, and structured logging.
- Instrumented agents with **Arize** (eval, drift, tracing) and **Datadog** (APM); deployed on **Kubernetes** and **OpenShift** via **GitHub Actions** and **Jenkins** CI/CD with SonarQube quality gates, using **Gradle** for JVM service builds.
- Deployed and debugged services in **Linux/Unix** environments (containers, OpenShift pods, VMs) using shell scripting, log analysis, and network troubleshooting for production support.
- Secured platform access with **AD/SSO**, **RBAC**, and **OAuth2**; partnered with executives on AI opportunity/ROI and go-to-market transformation.
- Mentored engineers on agentic system design, prompt engineering, and AI guardrails; owned end-to-end delivery with a 95% project success rate.
- Engineered scalable **frontend architecture** with **React.js**, **Next.js**, and micro-frontends; applied code-splitting, lazy loading, and dynamic imports to cut bundle size and improve TTI, optimizing platform performance by **60%**.
- Built a **reusable, configurable table component** adopted across 15+ modules, a **shared network layer** with consistent error handling and retries, and a **UI version synchronization system** (React + Nginx) so users always run the latest build.

Software Engineer —— Deqode Aug 2023 - May 2025

- Architected enterprise-grade **RAG pipeline** using **LangChain**, Pinecone/FAISS, and AWS, reducing LLM hallucinations by 32% and improving retrieval latency by 45% across 1M+ documents.
- Fine-tuned 7B/13B **Qwen3-VL** LLMs using **SFT**/**QLoRA** on **AWS SageMaker** GPU clusters, improving domain accuracy by 21% while cutting inference cost by 60%.
- Deployed production **LLM serving** on **GCP Vertex AI** with **vLLM**/TensorRT-LLM, serving 13B-parameter models at sub-200ms latency for 500K+ requests/day.
- Orchestrated cloud-native ML systems on **Kubernetes** with **MLflow** and Terraform; led platform modernization for 15+ ML teams, compressing deployment cycle from 3 weeks to 2 days.
- Deployed multi-agent conversational systems using LangChain Agents and AWS Bedrock API gateways; **mentored engineers** on RAG scaling and context-window optimization.
- Built **AI monitoring/observability dashboards** (Grafana, Power BI, Looker) tracking operational metrics, system health, and model drift.
- Built a **configuration-driven, schema-to-React renderer** supporting dynamic layouts, conditional fields, and versioned JSON configs for non-engineering UI rollout.
- Built Aadhaar-Voter ID linking UI in React using **Bootstrap**, scaling to **high-traffic production loads** with resilient, reusable components.
- Migrated **Redux** to modular **RTK** slices + RTK Query, reducing API overfetching by 78%.
- Developed editable virtualized grid (1M+ rows) using **react-window**, with smooth scroll and memoization.

- Led federated design **system adopted by 50+ teams**, enforced standards.
- Integrated Web Vitals + **Sentry** to track LCP, FCP, TTI across 500k+ sessions/month and optimize core web metrics.

Software Engineer —— Shiv Technolabs Feb 2023 - July 2023

- **Leveraged deep knowledge of browser internals** to optimize performance, systematically reducing reflow,repaint, and bundle sizes to achieve significant improvements in Core Web Vitals.
- **Implemented advanced state management** patterns across complex application workflows using MobX and modern ES6+ JavaScript features for predictable data flow.
- **Led frontend performance audits** analyzing JS engines compilation behaviors and execution contexts to eliminate memory leaks and runtime bottlenecks.
- Orchestrated the transition to **Next.js** methodologies, integrating **RESTful APIs** to achieve seamless data transfer across platforms, which decreased website bounce rate by **15%** and grew traffic.

Software Engineer —— Sourcebae May 2021 - Jan 2023

- Built full-stack applications with **React**, **TypeScript**, and **NestJS**, shipping REST API backends backed by **PostgreSQL** and **MongoDB** serving 100K+ users with sub-100ms p99 latency via Redis caching.
- Designed a reusable **TypeScript** component library with centralized state management, cutting feature delivery time by 40% across 5 concurrent product teams.
- Containerized services with **Docker** and established **CI/CD pipelines** with automated testing and staged rollouts, eliminating manual deployment errors and enabling zero-downtime releases.
- Built **offline-first UI** in React Native using SQLite for local data storage and background sync to update the server when reconnected.

PROJECTS

TradeBrains — Automated Real-Time Algorithmic Trading Platform

- Architected an end-to-end algorithmic trading platform predicting 20-minute stock movements by fine-tuning open-source generative models (**Llama 3.3 70B**) using 3-stage **RLHF** (**SFT**, **Reward Model**, **PPO**) and **GRPO** to align prediction outputs directly with target performance metrics.
- Implemented custom **GRPO/PPO reward functions** integrating multi-objective metrics (factual correctness, formatting compliance, and financial risk metrics) to eliminate hallucinations in high-stakes domain outputs.
- Engineered **QLoRA 4-bit** parameter-efficient fine-tuning (**PEFT**) pipelines for 8B-parameter sentiment models, reducing memory footprints while maintaining **88% structured score extraction accuracy**.
- Executed **Direct Preference Optimization (DPO)** and **KTO** fine-tuning on domain-specific instruction datasets, bypassing separate reward model latency while matching RLHF alignment quality.

EDUCATION

Rajiv Gandhi Proudyogiki Vishwavidyalaya Bhopal, India

Bachelor of Engineering (HONOURS) - Information Technology; GPA: 7.75 August 2016 - May 2020

ACHIEVEMENTS

- **AWS Certified Developer Associate**.
- **AWS Certified AI Practitioner**.
- **AWS Certified Cloud Practitioner**.
- **Hacker Rank Javascript Basic**.
- **Hacker Rank CSS certification**.
- **Sololearn React + Redux**

TECHNICAL SKILLS

- **LLMs & Agents**: LangChain, LangGraph, CrewAI, Google ADK, LlamaIndex, Hugging Face Transformers, OpenAI API, Ollama, vLLM, TensorRT-LLM, LiteLLM, MCP Servers, Agent Orchestration, RAG Systems, Prompt Engineering
- **Training & Optimization**: LLM Fine-Tuning (SFT, QLoRA, LoRA, PEFT), RLHF, GRPO, DeepSpeed, FSDP, Distributed Training, CUDA, Model Quantization (INT8, NF4)
- **ML & Deep Learning**: PyTorch, TensorFlow, scikit-learn, XGBoost, Classical ML, Feature Engineering, Model Optimization
- **Data Engineering**: Apache Spark (PySpark, Spark SQL), Databricks, Delta Lake, Kafka, Airflow, BigQuery, ETL/ELT, Feature Store (Feast)
- **Cloud**: AWS (SageMaker, EKS, Lambda, S3), GCP (Vertex AI, GKE, Dataproc, BigQuery), Azure
- **Backend Engineering**: Python, FastAPI, Flask, asyncio, SSE/WebSockets, Go, TypeScript, REST APIs, gRPC, Microservices, Redis, n8n
- **Databases & Vector Stores**: PostgreSQL, MongoDB, SQL Server, Oracle (PL/SQL), Redis, Qdrant, Milvus, Weaviate, Chroma, SQL
- **MLOps & Infrastructure**: MLflow, Kubeflow, Kubernetes, OpenShift, Docker, Terraform, Linux/Unix, Shell Scripting, Gradle, CI/CD (GitHub Actions, Jenkins, SonarQube), Model Monitoring & LLM Observability (Arize, Datadog), Drift Detection, A/B Testing
- **Security & Auth**: OAuth2, AD/SSO, RBAC

- **Frameworks:** Reactjs, Redux, Nextjs, SCSS, MaterialUI, ExpressJs, Jest, RTL, Enzyme
- **Tools:** Git/Github, Postman, Swagger, Figma, Sequelize ORM, Firebase, Nginx, Webpack, Babel, ESLint, Prettier
- **Other:** RESTful Web Services, Agile methodologies, Unit testing, Integration testing, micro frontends, Incident Response & Operational Excellence